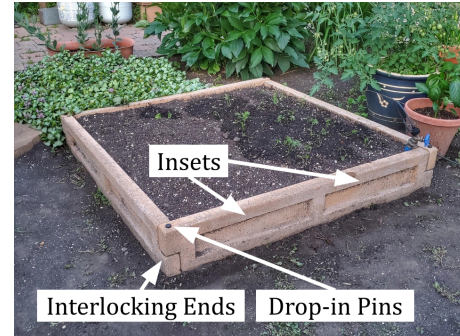


Modular Concrete Panels for Raised Garden Beds

Designer & Fabricator | Aug 2024 – Jul 2025

Overview:

Designed and fabricated a set of reusable molds for casting modular concrete panels used in raised garden beds. The panels featured interlocking pins for tool-free assembly, smooth rounded edges, and a decorative inset that reduced material usage and aligned with standard concrete bag volumes. The project combined hands-on fabrication, material planning, and iterative improvements based on mold performance, and casting efficiency.



Installed garden box assembled from precast concrete

Tools and Skills:

SketchUp, hand and power tools, mold design and fabrication, concrete casting and finishing, woodworking, project planning

Objective:

Design and fabricate reusable concrete molds to cast multiple panel sets for modular raised garden beds.

Design Highlights:

- Dimensional and finish lumber was selected for its availability and ease of fabrication for mold construction, balancing cost and practicality.
- Outward cambering of mold walls allowed easy release of castings without disassembly.
- Caulking was applied to internal corners of the mold to produce smooth, rounded edges on the panels.
- Concrete mix was dyed for long-lasting color that resists chipping, unlike surface stains.
- Interlocking pins ensured quick and straightforward assembly, with removable pipes placed in the mold to create the pin holes.
- Wire mesh reinforcement was embedded around thin edges near pin holes to improve durability.
- Insets improved aesthetics and reduced concrete volume, optimizing use of standard 60 lb bags.

Lessons Learned:

Wood proved unsuitable for concrete molds, as it warped and cracked panels during curing. Moisture-stable materials like vacuum-formed plastic or fiberglass would have been more durable despite higher upfront costs. Adding an outward camber to the mold walls eliminated the occasional need to disassemble molds between castings. Two molds proved sufficient, balancing workload and curing time better than the original four. A concrete mixing drum would have reduced manual effort and improved casting efficiency.

Next Steps:

Remake molds using a moisture-resistant material for improved durability and stability. Add a vibratory motor directly to molds to enhance concrete consolidation and reduce manual labor. Experiment with a smoother concrete mix to improve fluidity without increasing water content.

References:

Gravelle, K. (MANaboutTOOLS). *YouTube channel with concrete garden box tutorials*. YouTube channel.
<https://www.youtube.com/@MANaboutTOOLS>